

Architecture

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Architectural standard: modular, contract-bounded components

MessageFoundry is built as a **modular, loosely-coupled architecture with contract-defined boundaries (information hiding)** — so components can be built in parallel, by separate people or AI agents, without conflicts. This is the governing standard; the topology, store, and module map below are all instances of it.

Modular design (component-based architecture). The system is decomposed into independent components: the headless **engine**, the **code-first config** authored against it (Connections/Routers/Handlers + per-environment values), the **web console** (/ui), the **IDE extension**, the **PySide6 test harness**, and the **CLI tools** (generate / check / dryrun , beyond serve) — with the NSSM **Windows service** and **CI** as the operational wrapping. Within the engine itself: config / parsing / store / transports / pipeline / auth / api . Modularity bounds how much of the system any one change — or any one AI context window — has to hold, and deconflicts concurrent builds.

Information hiding (Parnas). The foundational principle is information hiding / encapsulation, from David Parnas's 1972 paper "On the Criteria To Be Used in Decomposing Systems into Modules." Its thesis is exactly our goal: decompose a system so each module hides a design decision behind an interface, **specifically so separate people can work on modules in parallel and a change inside one doesn't force changes in others**. That is the canonical citation for "split it up so teams don't conflict."

Separation of concerns — high cohesion + loose coupling. - **High cohesion** — everything related to one concern lives inside one component (the engine's message processing; the web console's UI). A change stays local. - **Loose coupling** — components depend on each other as little as possible, and only through stable seams. Low coupling means a change in component A has a small **blast radius**, so two people (or agents) editing A and B rarely touch the same code. - **Why it works:** parallel-work conflicts scale with **shared surface area**. Loose coupling minimizes that surface; high cohesion keeps each change confined to one module.

Contract-first / interface-driven design. Loose coupling only holds if the seams are explicit. We agree on the interface *first*, then each side builds behind it independently — the contract is the synchronization point; everything behind it is private and parallelizable. In this repo: - the **HTTP/WebSocket API** (api/app.py) is the contract between the engine and its clients (web console, IDE, harness); - the **connector registry** (transports/base.py) is the contract for pluggable transports; - the **one-way dependency rule** (pipeline / transports / parsing / store / config never import api) keeps the seams from leaking.

Organizational dimension. - **Conway's Law** — systems tend to mirror the communication structure of the teams that build them. The **Inverse Conway Maneuver** is the flip side: we draw module boundaries deliberately to match how we want teams (or AI agents) to split work — see the parallel git worktrees in **WORKTREES.md**. -

Module / boundary ownership — each component has one owner at a time (the DDD equivalent is a **bounded context**). Concurrent sessions keep their changes within their own module/branch.

Topology: engine-as-library + localhost API

The engine is an importable Python package (`messagefoundry`). Clients — primarily the **browser web console** (`/ui` , `messagefoundry_webconsole`) — drive it over a localhost HTTP + WebSocket API (FastAPI/uvicorn). The same API serves three deployments without code changes:

- **Embedded** — another Python program owns the engine in-process via `create_app(engine)` (the async test client; embedding). A **UI client is never this** — it's always a separate process (or a browser) that reaches the engine only over the HTTP API, never by importing it.
- **Local daemon** — engine runs as a Windows service / Linux daemon; the web console attaches over the API. See `SERVICE.md` for the Windows service setup (NSSM).
- **Remote** — same API over the network (Phase 2+, with auth/TLS).

We deliberately did **not** start with two separate processes + hand-rolled IPC. The logical boundary (library API) comes first; physical split is a deployment choice.

The same topology as a rendered diagram — clients are separate processes (or a browser) that reach the engine only through the API, and the engine packages never import `api` :

```
flowchart TB
    classDef client fill:#e3f2fd,stroke:#1565c0,color:#0d2b45;
    classDef api fill:#ede7f6,stroke:#5e35b1,color:#22103f;
    classDef engine fill:#e8f5e9,stroke:#2e7d32,color:#10240f;
    classDef deploy fill:#eceff1,stroke:#546e7a,color:#1c2429;

    CON["Web console /ui<br/>(browser)"]:::client
    IDE["VS Code extension"]:::client
    HARNESS["Test harness<br/>(PySide6)"]:::client

    subgraph API_BND["API — localhost 127.0.0.1 · auth + RBAC · the only external surface"]
        API["api/ — FastAPI + uvicorn<br/>HTTP + WebSocket"]:::api
        AUTH["auth/ — authn + RBAC<br/>deny-by-default · hash-chained audit"]:::api
    end

    subgraph ENGINE["Engine — headless asyncio service (no GUI imports)"]
        PIPE["pipeline/ — RegistryRunner<br/>listener · router · transform · delivery workers"]:::engine
        TRANS["transports/ — connector registry<br/>MLLP · File · X12 · DICOM C-STORE SCP (TCP/HTTP/DB planned)"]:::engine
        PARSE["parsing/ — pure HL7/X12/DICOM library<br/>python-hl7 · hl7apy · X12 codec · DICOM codec · base64 binary codec"]:::engine
        STORE["store/ — staged queue<br/>SQLite WAL · SQL Server · AES-256-GCM"]:::engine
        CFG["config/ — code-first wiring<br/>Connections · Routers · Handlers · environments/"]:::engine
    end

    NSSM["NSSM Windows service<br/>(messagefoundry serve)"]:::deploy

    CON -.->|"HTTP/WS API client"| API
    IDE -.->|"HTTP"| API
```

```

HARNESS -->|"MLLP send/receive"| TRANS
CON -->|"may import (pure lib)"| PARSE

API --> AUTH
API ==>|"depends on engine"| PIPE

PIPE --> TRANS
PIPE --> PARSE
PIPE --> STORE
PIPE --> CFG
TRANS --> PARSE

NSSM ==> ENGINE

```

The message store is the queue

The single most important reliability decision. We use a transactional **staged queue** on SQLite (WAL mode) — one generic **queue** table with a **stage** discriminator (**ingress** | **routed** | **outbound**). This is **Step B** of the staged pipeline (ADR 0001 — [docs/adr/0001-staged-pipeline-architecture.md](#)), the full router/transform split:

```

inbound msg → decode / parse / (strict-validate)  [synchronous – still NAK on failure]
|
▼ persist raw to the INGRESS stage → commit → ACK source  [ACK-on-receipt]
|   (ingress + routed lanes key on channel_id; outbound on destination_name)
▼ router worker (per inbound): run Router
produce one ROUTED row per selected handler + complete the ingress row  (one txn)
|
▼ transform worker (per inbound): run that handler's transform
produce an OUTBOUND row per destination + complete the routed row  (one txn)
|
▼ per-destination delivery worker
deliver → mark done | mark failed+reschedule (retry policy)

```

The **ACK is sent on receipt** — once the raw message is durably committed to the ingress stage, *before* routing/transform/delivery — so a slow or hung router/transform/outbound can no longer stall intake (the bottleneck this removes). Each stage **handoff** is a single committed transaction (claim → produce-next-stage rows → complete-this-stage), so a message is never lost or partially handed off; a crash before commit rolls the stage back and it re-runs (each handoff is idempotent against the re-run — the consumed row is gone), and **reset_stale_inflight** recovers in-flight rows of **every** stage on startup. This gives **at-least-once delivery**, **retries**, and **replay** without a separate broker. Because at-least-once now leans on a re-run re-deriving identical output, **routers and transforms must be pure**; **outbound connections must be idempotent**. Correlation IDs are persisted for de-duplication. Each destination drains independently — a slow/failed one never blocks siblings, and a slow transform never blocks routing.

Disposition flows with the message, decided by the store finalizer (count-and-log): **RECEIVED** at ingress → **ROUTED** / **UNROUTED** after the router → **PROCESSED** (all delivered) / **FILTERED** (every handler ran, delivered nothing) / **NOT_DEPLOYED** (every destination the handlers addressed is in the graph but **deployed=false**) / **ERROR** (dead-lettered at any stage) once nothing is still in flight. The finalizer is the **single authority** — it never finalizes

while any earlier-stage row is pending, so a delivered handler can't mark a message done while a sibling handler's routed row still awaits transform. The ACK means *receipt-and-persistence*, not a final disposition: a routing/transform failure is post-ACK, so it is logged + dead-lettered (operators rely on the disposition + AlertSink), not NAK'd.

NOT_DEPLOYED (BACKLOG #233, **ADR 0111**) is the one disposition the finalizer cannot read off the queue rows: it decides **FILTERED by absence** (no rows left, message still **ROUTED**), so a declined destination would otherwise collapse to **FILTERED** and misreport operator intent. The decline is therefore persisted where the finalizer can see it — the **transform worker** drops a **Send** whose target outbound is **deployed=false** *before* it becomes an outbound-stage row, and writes one **not_deployed message_events** row per declined destination in that same handoff transaction (**MessageStore.transform_handoff**). Zero deliveries **with** that event finalize **NOT_DEPLOYED**; without it, **FILTERED**. A deployed sibling that delivers still finalizes **PROCESSED**, with the event row carrying the skipped leg. That event is in the store's audit floor, so the **message_events** verbosity gate can never thin it away — dropping it would take the disposition with it. Dry-run and the Test Bench never run the finalizer, so a fully-declined message previews there as **FILTERED**: the declined names stop at **RouteOutcome.declined** and are not carried into **DryRunResult**.

The router/transform split was taken after Step A's measured write amplification (now ~3 durable transactions/message for a single-handler message; +1 per extra handler) — recorded in [docs/benchmarks/step-b-write-amplification.md](#). An optional per-inbound **ack_after=delivered** (defer the ACK until delivery succeeds) is planned but not built — the pipeline is ACK-on-receipt only.

The same staged flow as a rendered diagram:

flowchart TB

```
classDef stage fill:#fff3e0,stroke:#ef6c00,color:#3a1d00;
classDef worker fill:#e8f5e9,stroke:#2e7d32,color:#10240f;
classDef disp fill:#ede7f6,stroke:#5e35b1,color:#22103f;
classDef io fill:#e3f2fd,stroke:#1565c0,color:#0d2b45;

SRC(["Inbound connection<br/>MLLP / File"]>:::io
LISTEN["Listener<br/>decode · parse · (strict-validate)"]:::worker
NAK["NAK (AR/AE) + ERROR<br/>synchronous, pre-ingress"]:::disp

ING["ingress stage<br/>raw committed"]:::stage
ACK(["ACK (AA) – on receipt"]>:::io
RW["Router worker (per inbound)<br/>run @router – pure"]:::worker
ROUTED["routed stage<br/>one row per selected handler"]:::stage
TW["Transform worker (per inbound)<br/>run @handler transform – pure"]:::worker
OUT["outbound stage<br/>one row per destination"]:::stage
DW["Delivery worker (per outbound)<br/>idempotent send · retry · dead-letter"]:::worker
DEST(["Outbound connection(s)"]>:::io

FIN["Store finalizer<br/>single disposition authority"]:::disp
D1["RECEIVED"]:::disp
D2["ROUTED / UNROUTED"]:::disp
D3["PROCESSED / FILTERED<br/>NOT_DEPLOYED / ERROR"]:::disp

SRC --> LISTEN
LISTEN -->|"decode/parse/validate fail"| NAK
```

```

LISTEN -->|"ok"| ING
ING --> ACK
ING ==>|"committed txn"| RW
RW ==>|"committed txn"| ROUTED
ROUTED ==>|"committed txn"| TW
TW ==>|"committed txn"| OUT
OUT --> DW
DW --> DEST

ING -. ->|"records"| D1
RW -. ->|"records"| D2
DW -. ->|"records"| D3
D1 -. -> FIN
D2 -. -> FIN
D3 -. -> FIN

```

Concurrency

asyncio core. One listener + a **router worker** + a **transform worker** per **inbound connection**; one delivery worker per **outbound connection**. Listeners (MLLP/TCP), pollers (file/DB), the router/ transform workers, and retry timers are all asyncio tasks supervised by the `RegistryRunner` so a crash in one is isolated (a crashed worker is respawned; the router/transform workers drain their inbound's already-ACKed messages even while the source is stopped). Each worker class has its own wake event, so a producer wakes only its downstream consumer (no cross-class lost wakeups).

That isolation now extends to **startup wiring** (ADR 0031): a single connection that fails to build/bind at start (unresolvable `env()` /cert, an egress refusal, a port in use, a cleartext-exposure refusal) is recorded `failed` + logged + alerted, and the engine **starts the rest of the graph and serves the API** instead of aborting. A failed outbound still gets its delivery worker, so rows routed to it are **retried, never dropped**, and a reload/restart that builds it self-heals the lane; a failed inbound just isn't listening. Reload itself stays fail-fast (it `build_check`s the whole new registry before quiescing a healthy graph) — only startup degrades.

Parsing: tolerant-first, strict-on-demand

- `python-hl7` parses tolerantly and powers field *peek* for routing. Hot path.
- `hl7apy` does version-aware + profile validation, opt-in per inbound connection (`validation.strict`). Slower; kept off the routing hot path.

Real-world HL7 v2 is frequently non-conformant. Strict-by-default would drop messages that must still route, so tolerance is the default and strictness is an opt-in.

Configuration: connections + code-first routing

The target model is a **graph wired by name, authored as Python** — no enclosing "channel" object:

- a **Connection** is a named inbound or outbound endpoint (MLLP, file, ...);

- an inbound Connection names a **Router** — a Python script that sees every received message, decides which **Handler(s)** to forward to, and may filter;
- a **Handler** is a Python script that filters → transforms → sends to outbound Connection(s).

Config is version-controlled, diff-able Python; the database stores runtime state and messages **only**, never configuration (the opposite of Mirth burying channel XML in the DB).

Connections/Routers/Handlers are authored against the `messagefoundry` surface

(`inbound` / `outbound` / `@router` / `@handler` / `Send` / `MLLP` / `File` / `Message`); a directory of such modules loads via `load_config` into a `Registry` that the engine's `RegistryRunner` runs.

The configuration graph wired by name (no enclosing "channel" object) as a rendered diagram:

```
flowchart LR
    classDef conn fill:#e3f2fd,stroke:#1565c0,color:#0d2b45;
    classDef router fill:#fff3e0,stroke:#ef6c00,color:#3a1d00;
    classDef handler fill:#e8f5e9,stroke:#2e7d32,color:#10240f;

    IB["inbound: IB_ACME_ADT<br/>(MLLP)"]:::conn
    R["@router<br/>sees every message · filters · forwards by name"]:::router
    H1["@handler: to_EHR<br/>filter → transform"]:::handler
    H2["@handler: to_archive<br/>filter → transform"]:::handler
    OB1["outbound: OB_EHR_ADT<br/>(MLLP)"]:::conn
    OB2["outbound: OB_ARCHIVE<br/>(File)"]:::conn

    IB -->|"names a router"| R
    R -->|"forward to handler(s)"| H1
    R --> H2
    H1 -->|"Send → outbound"| OB1
    H2 -->|"Send → outbound"| OB2
```

PHI / security

Messages contain PHI. Access control and the *data* protections are tracked separately — see [SECURITY.md](#) (identity, RBAC, action audit) and [PHI.md](#) (data-at-rest, transport, logging, retention, de-identification), which is the authoritative built-vs-planned map. The per-interface trust boundaries and STRIDE threats are in [security/THREAT-MODEL.md](#) (PW.1–2 / ASVS V15), a maintainer-internal document — [SECURITY-DOCS-POLICY.md](#) explains what is withheld and what you can request.

Built today: authentication + RBAC (PHI views gated by `messages:view_raw` / `messages:view_summary`), a user-attributed append-only **audit log** (hash-chained) of who viewed/searched/replayed messages, and **encryption-at-rest** for message bodies — AES-256-GCM through the store cipher when a key is set, with owner-only DB/WAL file permissions and required volume encryption covering the rest.

Also built (see [PHI.md](#)): **PHI redaction on the logging path** — stdlib `logging` only (deliberately not structlog), with an always-on handler filter plus the `safe_exc()` chokepoint, so an exception rendered into a stored disposition or a log line is scrubbed of HL7-shaped content; **MLLP-over-TLS** (TLS 1.2+, server-certificate

verification with hostname checking, opt-in mTLS) configured per connection, with an off-loopback plaintext listener refused at startup; and **retention/purge** enforcement, run by the engine's async `RetentionRunner`.

Module map

Package / module	Responsibility
<code>messagefoundry.config</code>	Connector models (<code>models.py</code>) + code-first wiring registry/loader (<code>wiring.py</code>) + service settings (<code>settings.py</code>)
<code>messagefoundry.parsing</code>	Tolerant peek (python-hl7) + strict validate (hl7apy); parse tree (<code>tree.py</code>) and the <code>Message</code> transform model (<code>message.py</code>); pure non-HL7 codecs — X12 EDI (<code>x12/</code>), DICOM headers/SR over pydicom (<code>dicom/</code> , ADR 0025), and the base64 binary-carriage codec (<code>binary.py</code> , ADR 0028)
<code>messagefoundry.store</code>	Durable message store / staged queue (one <code>queue</code> table, <code>stage</code> = ingress outbound), SQLite WAL; every receipt logged with a disposition that flows with the message. <code>Store</code> protocol + <code>open_store</code> factory in <code>base.py</code> ; production SQL Server backend in <code>sqlserver.py</code>
<code>messagefoundry.transports</code>	Inbound & outbound connections (MLLP, file, X12 raw-TCP, DICOM C-STORE SCP inbound over pynetdicom — ADR 0025, ...), resolved through a registry (<code>base.py</code>) — never special-cased in <code>pipeline/</code>
<code>messagefoundry.anon</code>	Deterministic, secret-per-dataset pseudonymization / de-identification (fail-closed; ADR 0030) — exposed to the tee (<code>anonymize-captures</code>) and the test harness
<code>messagefoundry.pipeline</code>	Per-message routing/handling (<code>RegistryRunner</code> in <code>wiring_runner.py</code>) + per-inbound-connection supervision (<code>engine.py</code>); offline <code>dryrun.py</code>
<code>messagefoundry.api</code>	Localhost FastAPI surface (<code>app.py</code> + response <code>models.py</code>) — the engine's only external interface. The <code>/ui</code> browser web console mounts onto this app from a separate wheel (see below) via <code>mount_ui</code> , pinned against the <code>api/_ui_seam.py</code> contract (<code>WEBCONSOLE-PACKAGE.md</code>)
<code>messagefoundry.apiclient</code>	Qt-free / FastAPI-free engine-client library (ADR 0088) — a small typed httpx wrapper over the API, shared by the test harness and any future client
<code>messagefoundry.auth</code>	Authn + RBAC core (no FastAPI): permissions/roles, <code>Identity</code> , password hashing, opaque session tokens, LDAP/Kerberos (<code>service.py</code>) — enforced by <code>api</code>
<code>messagefoundry.generators</code>	Conformant synthetic HL7 generators (ADT, ORM, ORU, ...) behind <code>messagefoundry generate</code>
<code>messagefoundry.service</code>	Local Windows service control (<code>messagefoundry service {install,start,stop,status}</code> ; ADR 0088), wrapping the NSSM scripts
<code>__main__.py</code> , <code>checks.py</code>	CLI entrypoint (<code>serve</code> / <code>generate</code> / <code>check</code> / <code>service</code>) + the <code>check</code> commit/CI gate

Dependency direction is one-way: `pipeline` / `transports` / `parsing` / `store` / `config` never import `api` . The API depends on the engine; the clients (web console, harness) depend on the API (via `apiclient`). The former PySide6 desktop console (`messagefoundry.console`) was retired — its reusable Qt widgets moved to the harness (BACKLOG #103, ADR 0032 retired).

Beyond the engine packages (separate components, not `messagefoundry.*`): the **code-first config** an operator authors (their `--config` modules + `environments/` value overrides), the **web ops console** (`messagefoundry_webconsole/`, a separately-versioned second wheel `messagefoundry-webconsole` mounted same-origin under `/ui` — ADR 0065, [WEBCONSOLE-PACKAGE.md](#)), the **IDE extension** (`ide/`), the **test harness** (`harness/`), the sample config + message corpus (`samples/`), the **Windows service** scripts (`scripts/service/`), and **CI / supply-chain** (`.github/workflows/`).

Dependencies

Declared in `pyproject.toml` (the source of truth) as `>=` minimums; the pinned, hashed resolution lives in the committed `uv.lock` / `requirements.lock` (DEP-1, see [SECURITY.md](#)). Requires **Python 3.14+**.

Runtime

- `hl7` (`python-hl7`) — fast, tolerant parsing for the routing hot path
- `hl7apy` — version-aware validation + profiles (opt-in strict path)
- `pydantic` — config / connector models and validation
- `aiosqlite` — async SQLite for the message store / queue
- `fastapi` + `uvicorn[standard]` — localhost engine API
- `argon2-cffi` — argon2id password hashing
- `cryptography` — AES-256-GCM at-rest encryption for the store
- `ldap3` + `pyspnego` — Active Directory / LDAP auth and Windows SSO (Kerberos)

Optional extras

- `harness` → `PySide6` (LGPL — chosen so the OSS test harness GUI is distributable; not PyQt) + `httpx` + `truststore` (was `[console]` before the desktop console was retired — BACKLOG #103)
- `sqlserver` → `aiodbc` (production SQL Server store; also needs the OS-level Microsoft ODBC Driver 18 for SQL Server, which is not pip-installable; lazy-imported so SQLite-only installs skip it)
- `dicom` → `pydicom` + `pynetdicom` (DICOM codec — headers/SR only, no numpy — and the C-STORE SCP inbound connector; ADR 0025; lazy-imported so non-DICOM installs skip it)
- `dev` → `pytest`, `pytest-asyncio`, `httpx` (ASGI test client for the API), `ruff`, `mypy`

Build / tooling — `hatchling` (build backend), `Ruff` (format + lint, no Black), `mypy` (strict), `pytest`.